

ELITE FOUR CAPITAL

Political Event & Relative-Value Desk

Pricing, Arbitrage, Fees, and Timing in Binary Political Prediction Markets

A quantitative framework, applied to the 2028 Jon Ossoff complex

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1. Executive summary

A binary prediction-market contract pays \$1 if an event occurs and \$0 otherwise, so its price is a tradable estimate of the event's probability. This paper formalises how the desk prices, cross-checks, sizes, and executes these instruments, and where fees, venue choice, and settlement timing change the answer. The worked example is the cluster of markets on Senator **Jon Ossoff** for 2028; all figures are live as of 16 July 2026.

Five edges, in decreasing cleanliness:

- **A hard no-arbitrage constraint.** Winning the presidency requires the nomination, so $P(\text{presidency}) \leq P(\text{nomination})$ always. A monitor flags any violation as a locked arbitrage or a data error.
- **A cross-venue lock.** The same contract prices differently on Polymarket and Kalshi. Long the cheaper venue's Yes and short the richer venue's No pays \$1 in every state for less than \$1 — a genuine arbitrage, net of fees and two-platform frictions.
- **A Dutch book on the overround.** Kalshi's candidate field sums to ~105%; buying No on every name locks ~5% minus fees. Polymarket links its field to ~100%, so no Dutch book there.
- **A relative-value wedge.** The two markets imply a conditional general-election win of ~52–60% if nominated — rich for a first-term senator. Long nomination / short presidency isolates the “nominated-but-loses” state cheaply.
- **A timing edge.** The nomination is decided on the primary calendar and re-rates to ~99¢ months before the November 2028 settlement — harvest the move, don't hold to term.

Overriding caveats: all markets load on one factor (Ossoff's standing); edges are perishable (the Ossoff cross-venue gap reversed intraday on 16 July); and venue choice trades price against fees, tax cleanliness, and counterparty safety.

2. Pricing framework

For event E , a \$1-on-Yes contract trading at $q \in (0,1)$ satisfies, absent fees and time value, $q = \mathbb{P}(E)$, the (risk-neutral) probability. Denote the desk's fair estimate p ; the signed edge on a long-Yes is $p - q$. Live inputs: nomination $q_N \approx 0.14$, presidency $q_P \approx 0.075$, horizon $\tau \approx 2.31$ years.

3. The no-arbitrage subset constraint

Because $P \subseteq N$ (Ossoff reaches the presidency only as nominee):

$$q_P \leq q_N.$$

If violated, buy Yes-nomination and No-presidency: the pair returns $\geq \$1$ in every state, so any cost below \$1 is riskless. The desk monitors this continuously as both a mispricing alarm and a data-integrity check — a stale feed once inverted it.

4. Conditional decomposition and the wedge

By definition $q_P = q_N \cdot c$, so the market's implied conditional is $c = q_P/q_N \approx 0.52$ – 0.60 . For a first-term swing-state senator this is demanding (generic out-party nominee ≈ 0.50 , lesser-known candidates run behind). The **wedge** — long nomination, short presidency — isolates the difference set $N \setminus P$ (“nominated but loses”), priced at $w = q_N - q_P \approx 5\text{¢}$ against a modelled fair of $p_N(1 - \hat{c}) \approx 6$ – 8¢ . Its residual exposure is to c alone, and it is hedged against Ossoff's overall trajectory.

View sensitivity. If your estimate of c is *above* the market's (e.g. you believe a strong nominee rides an anti-incumbent wave to the White House), the wedge is the wrong trade — buy the presidency directly at $\sim 7\text{¢}$ for the largest multiple.

5. Cross-venue pricing: overround and fees

The identical nomination event trades richer on Kalshi than Polymarket. Two distinct effects:

- **Overround.** Kalshi lists each candidate as an independent market; the field sums to $\sim 105\%$ (a 5-point house margin), so every Yes runs rich. Polymarket's neg-risk linkage arbitrages the field to $\sim 100\%$. Live snapshot (16 Jul): Ossoff 12¢ (Poly) vs 15¢ (Kalshi), Beshear 2.1¢ vs 3.8¢.
- **Clientele lean.** On two-way partisan markets the gap flips by crowd: in 2024 Polymarket ran 5–8 points hotter for Trump than Kalshi.

Fees. Both charge a taker fee of the form $\text{coeff} \times \text{price} \times (1 - \text{price})$; Kalshi's coefficient is 0.07, Polymarket's 0.04. Makers pay zero. As a share of stake the fee is $\text{coeff} \times (1 - \text{price})$ — worst on cheap longshots:

Contract price	Kalshi fee (of stake)	Polymarket fee (of stake)
3¢ (deep longshot)	6.8%	3.9%
14¢ (Ossoff)	6.0%	3.4%
50¢ (toss-up)	3.5%	2.0%
90¢ (favorite)	0.7%	0.4%

All-in optimiser. Combining price and fee into an all-in cost $q \cdot (1 + \text{coeff} \cdot (1 - q))$, Polymarket usually wins for buying Yes on candidate longshots — but the edge is perishable. On 16 July, Ossoff's Polymarket price rallied 12¢ $\rightarrow$ 14.4¢ and the optimal venue flipped from Polymarket (all-in 12.4¢) to Kalshi (14.8¢) within the hour.

6. The arbitrage scan

Cross-venue lock. Buy Yes on the cheaper venue and No on the richer venue. One leg pays \$1 in every outcome, at a cost of $q_{\text{cheap}} + (1 - q_{\text{rich}})$, locking $q_{\text{rich}} - q_{\text{cheap}}$ minus the two taker fees. It is a true arbitrage in payoff terms; the frictions are separate collateral on two platforms, fees, US-access rules, and the requirement that both venues resolve identically (same event, same source).

Dutch book on the overround. With the Kalshi field at $\sim 105\%$, buying No on all N candidates costs $\sum (1 - q_i) = N - 1.05$ and pays \$1 on the $N - 1$ losers, for $\sim 5\%$ gross minus fees. Risks: incomplete fills across illiquid names, and a write-in / late entrant. Polymarket's $\sim 100\%$ field offers no such book.

These are surfaced live on the dashboard's Arb tab; all are net-of-fee and flagged only when the edge clears frictions.

7. Settlement timing — the convergence clock

The nomination contract settles in November 2028 but is economically decided on the primary calendar (Feb–Jun 2028), converging to $\sim 99\text{¢}$ by the August convention. The desk therefore harvests the re-rating and exits early rather than holding to term — collapsing a 2.3-year lock into weeks. The presidency contract does **not** converge early; it remains live to Election Night. A withdrawal announcement collapses the nomination Yes to a few cents in one step.

8. Position sizing (Kelly)

For a binary at price q with fair p , net odds $b = (1 - q)/q$ and the growth-optimal fraction is $f^* = (p - q)/(1 - q)$. At $q = 0.14$, $p = 0.15$, $f^* \approx 1.2\%$ of bankroll at full Kelly; the desk sizes at **quarter-Kelly**. Because the three markets are highly correlated, the effective fraction is reduced further at the book level.

9. Time value and the hold-vs-trade decision

Collateral earns no yield, yet resolution is ~ 2.3 years out. Expected annualised return if held is $(p/q)^{(1/\tau)} - 1 \approx 6-7\%$, only $\sim 2-3\%$ above cash and attached to a lumpy binary payoff — unattractive risk-adjusted. Combined with \$7, the conclusion is to **trade the re-rating, not hold to resolution**; Polymarket's continuous order book makes early exit routine.

10. Regulation, counterparty, and payout

Dimension	Kalshi	Polymarket
Counterparty	CFTC-regulated clearinghouse (Kalshi Klear) guarantees payout	Smart contract + crowd; an oracle (UMA) resolves
Your funds	Segregated, bankruptcy-remote USD (JPMorgan, BNY Mellon)	Self-custodied USDC; no segregation, no insurance
Shutdown risk	Lower — federally regulated; won its CFTC fight	Mixed — geo-block history + state actions (MN, NV)
Tax record	USD; limited 1099s; cleaner	USDC = every trade also a crypto disposal

Neither is FDIC/SIPC insured. The $\sim 2-3\%$ Kalshi price premium is, in effect, a **safety-and-tax tax** — you pay more for a guaranteed payout and cleaner records. Whether it is worth paying depends on size and holding period.

11. Risk model and recommended positions

- **Single-factor concentration.** All markets load on Ossoff's standing; treat as one factor, cap $\leq 5\%$ of bankroll.
- **Perishability.** Edges converge fast (the 16 July reversal); enter on dislocation, exit on re-rating.
- **Data quality.** Per-market endpoints with timestamps; the aggregate cache was found 30 days stale and is not trusted for execution.

Trade	Rationale	Sizing
Long nomination (cheaper venue)	Overround + fee edge on Polymarket; harvest the primary re-rating.	$\frac{1}{4}$ -Kelly
Wedge: long Nom / short Pres	Fades a rich $\sim 52-60\%$ conditional; hedged to Ossoff's path.	Primary RV
Cross-venue lock / Dutch book	When the Arb tab shows a net-of-fee edge and both venues are accessible.	Opportunistic
Buy presidency (only if bullish)	For a "wins it all" view — biggest multiple, cheapest on Polymarket.	Directional

Appendix — key formulas & sources

- Price = probability; edge = $p - q$. No-arb: $q_p \leq q_n$. Conditional $c = q_p/q_n$.
- Wedge price $q_n - q_p$; fair $p_n(1 - \hat{c})$. Kelly $f^* = (p - q)/(1 - q)$, $\times \frac{1}{4}$.
- Taker fee = $\text{coeff} \cdot q \cdot (1 - q)$, coeff 0.07 (Kalshi) / 0.04 (Poly); all-in $q \cdot (1 + \text{coeff} \cdot (1 - q))$.

- Cross-venue lock = $q_{\text{rich}} - q_{\text{cheap}} - \text{fees}$. Dutch book = $\sum (1 - q_i)$ vs $(N - 1)$ payout = overround - fees.
- Annualised IRR if held = $(p/q)^{(1/\tau)} - 1$; time-value hurdle $(1 + r_f)^\tau$.

Sources: Polymarket gamma API and Kalshi trade API (live, 16 Jul 2026); Kalshi and Polymarket fee schedules; Sherwood/CFTC on 2024 cross-venue divergence. Educational research; not investment advice.